

Comparative Analysis of Machine Learning Algorithms for HIV/AIDS Prediction

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Abstract—Early detection of HIV/AIDS is important in order to reduce transmission rate. By evaluating different machine learning algorithms, this research identifies the best algorithm or model for HIV/AIDS detection. Even though there are previous research that has shown good results, they still have some challenges to be solved. This study addresses the problems of model interpretability by using an Explainable AI called SHAP in order to improve model interpretability and transparency.

Keywords—machine learning, HIV/AIDS, early detection, explainable AI, healthcare

I. INTRODUCTION

Human Immunodeficiency Virus (HIV) and Acquired Immune Deficiency Syndrome (AIDS) are still recognized as crucial global health challenges, affecting millions of people worldwide [1, 2]. Human Immunodeficiency Virus was first identified in the early 1980s and it attacks the body's immune system, specifically targeting CD4+ T cells, which are crucial for maintaining a functional immune response [1, 3]. HIV/AIDS is an infectious disease and the primary modes of HIV transmission are unprotected sexual intercourse, vertical transmission from mother to child during pregnancy, exposure to contaminated hypodermic needles, and transfusions involving infected blood [4]. Despite significant advancements in medical research, HIV/AIDS remains as a chronic disease that affects a lot of people globally. One of the current treatments for HIV/AIDS patients is called antiretroviral therapy (ART) and it has proven to enhance the patient's life expectancy [5]. However, HIV/AIDS sufferers require lifelong treatment as there is no definitive cure for this disease [5, 6].

As a result of no definitive cure discovered to treat HIV/AIDS, early detection of this disease is very important. Early detection of this disease is important for proper diagnosis and infection management, ensuring that patients will be treated with a suitable and appropriate treatment. In addition, as HIV/AIDS can be transmitted through several media, early detection can contribute to prevent the transmission of HIV/AIDS disease [7].

Following the advancement of technology, the healthcare sector starts to utilize artificial intelligence, specifically machine learning in order to help doctors to

improve predictive accuracy, diagnostic efficiency, and clinical decision-making capabilities. In addition, machine learning algorithms can help doctors to identify early detection of a disease [8]. There have been a lot of studies that utilize machine learning in order to predict or detect disease, starting from common disease to chronic disease, such as influenza [9], diabetes [10], heart disease [11], and cancer [12]. This shows that machine learning is making a significant impact on healthcare and is becoming more important in the field of healthcare, especially in predicting or detecting diseases [13]. Furthermore, the utilization of machine learning models to predict diseases does not stop here. It keeps extending further to address other complex and chronic diseases, such as HIV/AIDS.

As this paper was written, there have been successful studies that are using machine learning to predict HIV/AIDS. There are studies that focus specifically on detecting HIV infection among men who have sex with men (MSM) [14, 15]. For example, one study found that Random Forest was accurate in predicting HIV among high risk groups (MSM) in Zhejiang, China [14]. Another study showed that Decision Trees were helpful in predicting adolescents HIV testing in Ethiopia [16]. Furthermore, there is also a paper that discusses the role of machine learning in HIV risk prediction with the integration of Explainable AI [17]. However, this paper only provides an overview, discusses the theoretical role, potential benefits of machine learning in HIV risk prediction, and the importance of model interpretability. The paper does not include model development, experiment results or accuracy measurements.

Therefore, it can be concluded that these studies still face challenges like model interpretability becoming an area for further research. This research paper aims to compare different machine learning algorithms in order to find out which one works best for detecting HIV/AIDS. By testing these methods, this study expects to overcome the existing challenges and figure out the best algorithm to use for HIV/AIDS detection. In addition, this study will also integrate Explainable AI in order to enhance the model interpretability.

II. RELATED WORKS

In the last few years, there has been a lot of interest in the use of artificial intelligence in healthcare, especially in the area of machine learning for disease detection or

prediction. Specifically, machine learning has become increasingly important for improving predictive accuracy, diagnostic efficiency, and decision-making capabilities in complex healthcare scenarios, such as HIV/AIDS detection.

Several studies have explored how machine learning can enhance healthcare outcomes compared to conventional statistical methods. For example, a research using machine learning models to analyze data from multiple surveys demonstrated that these models outperformed traditional statistical approaches in HIV/AIDS risk prediction [18, 19].

One significant area of research involves predicting HIV infection among high-risk groups. For instance, one study focused specifically on detecting HIV infection among men who have sex with men (MSM). The study evaluated several machine learning models including Logistic Regression, Decision Trees, Support Vector Machines (SVM), and Random Forest. The results indicated that the Random Forest model achieved the highest accuracy [14].

Researchers have also applied machine learning to improve HIV testing services, which focused on adolescent HIV testing in Ethiopia. This study used models such as Decision Trees, Random Forest, and Logistic Regression to identify factors that influence testing. The study found that the J48 Decision Tree model has a high accuracy rate and is the best performing algorithm. The study also identified important factors/crucial predictor features, which are age, knowledge of HIV testing locations, age at first sexual encounter, recent sexual activity, and exposure to family planning [16].

Another study examined how machine learning can classify HIV/AIDS cases using clinical data. The researchers tested different machine learning models, including Gradient Boosting and Naive Bayes, on a dataset of 50,000 samples. The results showed that while Gradient Boosting achieved the highest overall accuracy, Naive Bayes was better at detecting positive cases. This shows the importance of balancing different models to ensure both accuracy and sensitivity in medical diagnosis [20]. In addition to predictive modeling, there are also studies that explored methods for identifying socio-behavioral factors associated with HIV/AIDS risk. A study using Gradient Boosting Trees and SVM analyzed data from East and Southern Africa and achieved high predictive performance. The results showed the significance of using socio-behavioral characteristics to enhance HIV screening and policy development [21].

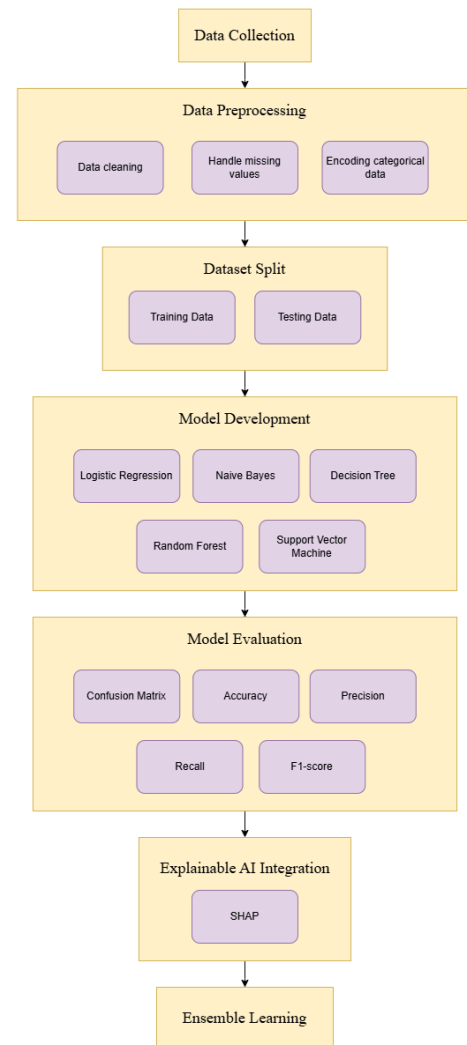
Moving to the integration of Explainable AI, there is a paper that explores the role of machine learning in HIV risk prediction with the integration of Explainable AI [17]. This paper focused on providing insights about the theoretical role, potential benefits of machine learning in HIV risk prediction, and the importance of implementing Explainable AI, such as SHAP to enhance the model interpretability. The gap of this paper is that it does not specifically conduct model development and experiments to predict or detect HIV/AIDS.

These existing studies collectively showed the growing role of machine learning in HIV/AIDS prediction. They demonstrated the effectiveness of different models in various contexts, from clinical diagnosis to behavioral risk assessment. However, despite the promising results, challenges such as model interpretability remain areas for further research. Addressing these problems could be important in advancing the real-world application of machine learning for HIV/AIDS detection and prevention.

In order to address these gaps, future experiments need to focus on comparing different machine learning models for HIV/AIDS detection or prediction with the integration of Explainable AI, such as SHAP to evaluate and enhance model interpretability.

III. METHODOLOGY

This study aims to compare the performance of different machine learning models (Logistic Regression, Naive Bayes, Decision Tree, Random Forest, and Support Vector Machine) in order to detect HIV/AIDS infection. This study also integrates Explainable AI, such as SHAP to enhance the model interpretability. The methodology of this study is divided and organized into 6 main stages: (a) data collection, (b) data preprocessing, (c) model development, (d) dataset split, (e) model evaluation, and (f) ensemble learning. Fig. 1 illustrates the flow of this study in order to achieve the research objectives.



A. Data Collection

The dataset used for this study is obtained from Kaggle [22]. This dataset contains 698 samples. Each data contains 10 features which are:

- **Age:** Numerical representation of the patient's age.
- **Marital Status:** Whether the patient is married, unmarried, divorced, etc.
- **STD:** The patient's history of sexually transmitted diseases (Yes/No).
- **Educational Background:** The patient's highest level of education.

- **HIV Test in Past Year:** Whether the patient took an HIV test in the last year (Yes/No).
- **AIDS Education:** Whether the patient has received education about AIDS (Yes/No).
- **Places of Seeking Sex Partners:** Locations where the patient seeks sexual partners.
- **Sexual Orientation:** The patient's sexual preference (e.g., heterosexual, homosexual, bisexual, etc).
- **Drug-Taking:** Whether the patient has a history of drug use (Yes/No).
- **Result:** The HIV/AIDS status (Positive/Negative).

This dataset is suitable for this research because the dataset provides relevant patients' socio-behavioral features that are commonly linked with HIV/AIDS. As a result, the machine learning models will be able to use these to accurately find the pattern and correlations to HIV/AIDS.

B. Data Preprocessing

In order to ensure that the machine learning models can perform optimally, the dataset used must be preprocessed and cleaned. During this stage, researchers are dealing with missing or null values in the dataset with several appropriate methods and techniques either by deleting the field or row with missing values or replacing the missing values using some strategies. For numerical data, the missing values can be replaced using the mean or median strategy. On the other hand, to deal with missing values in categorical data, several strategies like replacing the missing value with the most frequent value can be used. This process prevents the machine learning model from producing inaccurate results. Dealing missing values could be done using the SimpleImputer function from sklearn library. After dealing with missing values, the next step is to convert categorical data into numerical data, so that the data could be easier to be processed and studied by the machine learning models. Converting categorical data into numerical data could be done using the LabelEncoder function from sklearn library.

C. Model Development

1. **Logistic Regression**
Logistic Regression is a classification algorithm which categorizes data into groups. It predicts the result by using a sigmoid function. If the probability is above 50%, it belongs to a class. However, if it is below 50%, it belongs to the other class.
2. **Naive Bayes**
Naive Bayes is another classification algorithm based on probability and Bayes' theorem. It calculates the result by assuming that the features are independent and do not affect each other. It is called 'naive' because this assumption is often false.
3. **Decision Tree**
Decision Tree classifies data by simply splitting it based on yes or no questions

repeatedly. At each step, the algorithm picks the best feature to separate data clearly. It keeps on creating branches until there is a clear classification.

4. **Random Forest**
Random Forest is a classification algorithm that is like making many decisions and outputs the mode of the classes of the trees. By averaging multiple decision trees, it is more accurate than a single decision tree.
5. **Support Vector Machine**
Support Vector Machine classifies the data by finding the hyperplane that best divides a dataset. It focuses on maximizing the margin between data points of different classes. This makes it effective in high dimensional spaces.

D. Model Evaluation

The performance of machine learning models could be evaluated using a set of evaluation or performance metrics. The typical evaluation metrics that are used for evaluating the performance of classification models are confusion matrix, accuracy, precision, recall, and f1-score.

1. **Confusion Matrix**
Confusion matrix is a simple table that is used to evaluate classification models by comparing its predictions to the actual results. Confusion matrix decomposes the predictions into four categories: correct predictions for both classes (true positives and true negatives) and incorrect predictions for both classes (false positives and false negatives). Confusion matrix displays the correct and incorrect predictions and by using this data, it helps us to calculate the other metrics like accuracy, precision, recall, and f1-score.
2. **Accuracy**
Accuracy simply measures how often the classification models make the correct prediction. It is the ratio between the number of correct predictions and the total number of predictions.
$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$
3. **Precision**
Precision is a measure of correctness that is achieved in true prediction. It tells us how many predictions are actually positive out of all the total positive predicted. In addition, precision is a useful metric in cases where false positives are a higher concern than false negatives.
$$Precision = \frac{TP}{TP + FP} \quad (2)$$
4. **Recall**
Recall is a measure of actual observations which are predicted correctly. Recall is also known as sensitivity, and it is a useful metric in cases where false negatives trump false positives.
$$Recall = \frac{TP}{TP + FN} \quad (3)$$
5. **F1-score**
F1-score is a number between 0 and 1 and it is the harmonic mean of precision and recall. The f1-score maintains the balance

between precision and recall for classification models.

$$F1 - score = \frac{2TP}{2TP + FP + FN} \quad (4)$$

E. Explainable AI Integration

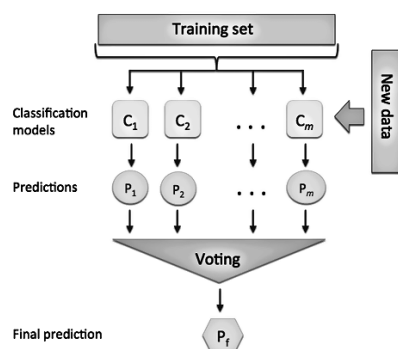
This study will apply an Explainable AI technique called SHAP (SHapley Additive exPlanations) to see how the machine learning algorithms predict HIV/AIDS. By doing so, it will provide transparency and give reasoning behind the results. SHAP will measure how much a feature contributes to the overall result.

The reason behind using SHAP is because it offers clear, detailed, and consistent explanations. Unlike other interpretability methods, it provides local (individual prediction) and global (overall model) explanations. Also, SHAP will show the effects of each feature, which will make the predictions easier to understand.

The result from using SHAP will then be used to determine which one plays the most significant role. The SHAP plots will be used to see which features are the most important in predicting HIV status. These visuals will also help understand how different features affect the model's decisions and how they interact with each other. This will be important in supporting the research by helping medical professionals recognize which are the most significant clinical indicators for HIV/AIDS diagnosis.

F. Ensemble Learning

Ensemble learning is a technique that combines the results of machine learning algorithms in order to improve the performance. By doing this, the group of 'weak learners' can create a 'strong learner' that is unlikely to be affected by small changes in the data. This study will incorporate ensemble learning to enhance HIV/AIDS prediction.



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